

Scientists investigated the effect that the release of heated water into a river from a power station had on the biodiversity of a local fish community over 29 years.

They measured the species richness and the number of fish of each species at the same site in October every year. The scientists used this information to calculate an index of diversity (d) of fish for each year.

Figure 7 and **Figure 8** show their results.

Figure 7

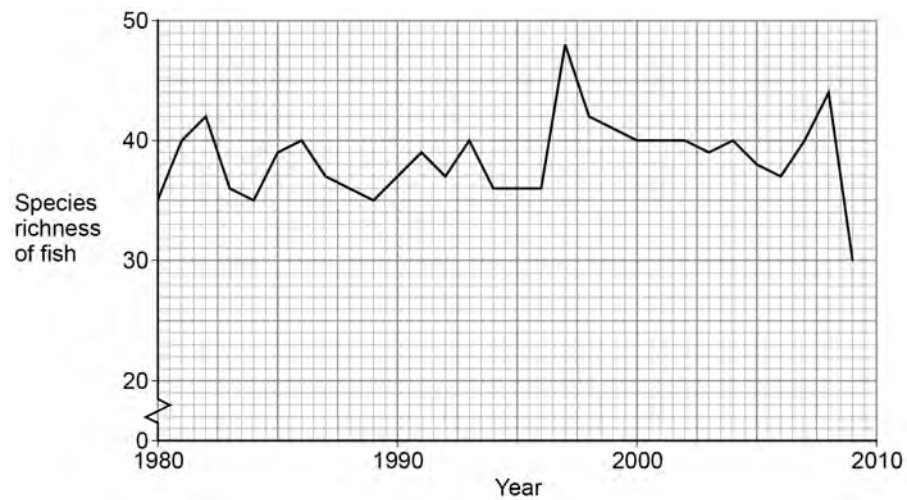
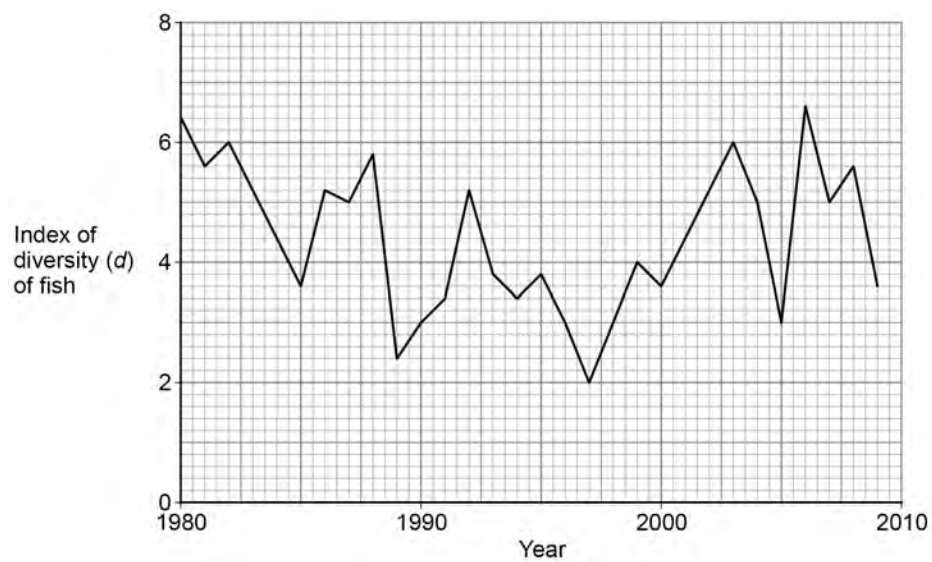


Figure 8



0	5	.	1
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The scientists used the following formula to calculate the index of diversity (d) of fish.

$$d = \frac{N(N-1)}{\sum n(n-1)}$$

where N = total number of fish of all species
and n = total number of fish of each species

In some years, the values were $N = 624$ and $\sum n(n-1) = 64\,792$

Which years had these values?

Use **Figure 8** and the formula above to work out your answer.

[1 mark]

Answer _____

0	5	.	2
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In 1997, the scientists recorded the highest species richness, but the lowest value of d over the 29 years.

Describe **and** explain how these results for 1997 were possible.

[2 marks]

Question 5 continues on the next page

A journalist studied **Figure 7** and **Figure 8** (on page 16) and concluded that releasing heated water from a power station has no effect on local fish communities.

Use all the information to suggest reasons why the journalist's conclusion might **not** be valid.

[4 marks]

[illegible]

Turn over for the next question

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